

Evaluate a detector

Evaluation matches predictions to labeled objects, counts distinct kinds of error, and reports metrics tied to an operating decision.

15–20 MINUTES

Prerequisite: boxes and IoU

METRICS EXERCISE

1 • Match

Pair predictions with ground truth.

2 • Count

TP, FP, and FN.

3 • Summarize

Precision, recall, and AP.

4 • Diagnose

Slice by deployment condition.

Start with matching

A prediction becomes a **true positive** only when it matches a ground-truth object under the evaluation rule—normally the correct class and sufficient IoU, with each ground-truth object matched at most once. An unmatched prediction is a **false positive**. An unmatched ground-truth object is a **false negative**.

This is why “the expected class appeared somewhere in the frame” is weaker than detection accuracy: it ignores localization, duplicates, other incorrect detections, and missed instances.

Central mental model

Evaluation is a matching procedure followed by counts. Metrics summarize those counts; they do not replace inspecting where the errors occur.

Precision and recall

$$\text{precision} = \text{TP} / (\text{TP} + \text{FP}) \quad \text{recall} = \text{TP} / (\text{TP} + \text{FN})$$

Precision asks: of the detections emitted, how many were correct? Recall asks: of the objects present, how many were found? Raising a score threshold often improves precision and reduces recall. The useful operating point depends on the cost of false alarms and misses.

Average Precision (AP) summarizes the precision–recall relationship across score thresholds for a class at a chosen IoU rule. COCO-style mAP averages more broadly across classes and several IoU thresholds. Always name the protocol; “mAP” alone is ambiguous.

Browser exercise: compute the metrics

An evaluation produces 8 true positives, 2 false positives, and 4 false negatives.

Compute precision and recall

Now ask what the aggregate hides: Were all four misses red cubes? Did they occur in low light? Were the false positives colored posters? Condition-level slices turn a score into diagnostic evidence.

Design the test before tuning

A test matrix should cover object classes, distances, viewpoints, lighting, occlusion, backgrounds, and empty scenes relevant to deployment. Record capture provenance. Freeze the evaluation set before using it to choose a threshold or model.

Applied to our cubes

MEASURED EVIDENCE The project has validation metrics and live test reports, but they answer different questions under different samples and procedures.

KNOWN LIMITATION Per-class frame-hit coverage is not ground-truth detection accuracy. It can be useful as a smoke signal while remaining unable to prove correct boxes or reject extra detections.

RECOMMENDED MATRIX Evaluate red, green, and blue cubes separately and together across room lighting, distances, orientations, clutter, and empty scenes. Report TP/FP/FN after class-and-IoU matching.

Project anchors: M5 live report and validation metrics.

Check your understanding

Which metric most directly captures missed objects?

- ☐ Recall across labeled objects
- ☐ Precision across emitted predictions
- ☐ Runtime across processed frames

Explain it back

Given TP, FP, and FN, explain precision and recall to a teammate. Then explain why one scalar metric cannot diagnose a room-domain failure.

Primary sources

- Microsoft COCO paper — detection dataset and evaluation setting.
- D2L: Object detection datasets.